

Chimwera T

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

MARKING SCHEME

JUNE 2015

CHEMISTRY

9189/5

1 (a) (i) Table 1:1

- ALL values recorded to 2 decimal places value within the range 2.45 g and 2.55 g, $3.46 - 3.54$ [1]
- The values recorded in appropriate places and correct subtraction; [1]

(ii) Table 1:2

- All values recorded to 2 decimal places $34.55 - 35.45$ [1]
- Correct subtraction; [1]
- Value within the range 34.50 cm³ and 35.50 cm³ last digit must be either zero / five. [1]

(iii) Table 1:3

- All values recorded to 2 decimal places and / at least two titre values with 0.1 cm³ difference; [1]
- No more than one zero start, correct subtraction; [1]
- Values within 0.1 cm³ difference ticked / average value calculation shown and summary completed; [1]

Enough titres with ± 0.1 cm³ [1] [8 marks]

one titre subtract 4

2 (a) Calculation of the difference using the following relationship

$$\frac{\text{weighed mass} / 5}{\text{supervisor's titre value}} - \frac{\text{weighed mass} / 5}{\text{candidate's titre value}}$$

$$\frac{\text{supervisor's dilution value}}{\text{supervisor's dilution value}} - \frac{\text{candidate's dilution value}}{\text{candidate's dilution value}}$$

1 titre max [8]

round up to 4 d.p.

Award accuracy marks from the difference calculated and spread penalty using the following table:

Accuracy table

Difference	Mark awarded
≤ 0.0020	12
0.0020+ to 0.0040	11
0.0040+ to 0.0060	10
0.0060+ to 0.0080	9
0.0080+ to 0.0100	8
0.0100+ to 0.0140	7
0.0140+ to 0.0180	6
0.0180+ to 0.0220	5
0.0220+ to 0.0280	4
0.0280+ to 0.0380	3
0.0380 to 0.0500	2
0.0500 to 0.1000	1
>0.1000	0

$$\text{Diff} = \left(\frac{\text{Wt mass} \times \text{titre}}{5} \right)_{\text{Sup}} - \left(\frac{\text{Wt mass} \times \text{titre}}{5} \right)_{\text{Cand.}}$$

[6 marks]

the smallest from biggest ticked. biggest — smallest (ticked)
 If not ticked/1
 3
 Spread Penalty Table
 +ve

Spread range/cm ³	Penalty
≤ 0,20	0
0,20+ to 0,30	1
0,30+ to 0,40	2
0,40+ to 0,50	3
0,50+ to 0,60	4
0,60+ to 0,70	5
0,70+ to 0,80	6
0,80+ to 0,90	7
0,90+ to 1,00	8
1,00+ to 1,10	9
1,10 to 1,20	10
1,20 to 1,50	11
> 1,50	12

(b) (i) $\frac{25,0}{1000} \times 0,098 / 0,025 \times 0,098 = 0,00245 \text{ moles}$ [1]

(ii) Answer to (b)(i) $\times \frac{1000}{\text{titre value}}$ [1] ECF (error carried forward) $8,75 \times 10^{-2}$
 mode ratio 1000 [1] / Eqn [1] balanced [1] / Rxn ratio [1]

(iii) Dilution volume $\times 2,0 \text{ moldm}^{-3} / 1000$ [1] n(HCl) that reacted =
 (c) (i) Answer to (b) (iii) - answer (b) (ii) $\times \frac{250}{100}$ [2] ans (b)(i) = [1]
 moles of HCl left as excess = ans (b)(i) $\times \frac{250}{25}$ [1]
 Answer (b) (iii) - answer to (b) (i) $\times \frac{250}{\text{titre value}}$ $0,0245$ [1] ans (b)(i) $\times \frac{250}{25}$

(ii) Answer to (c) (i) $\times \frac{1}{2} \times 106 = \text{Ans to working [1]}$ [2] $1,299 \times 1,3$
 Evaluation [1] $1,299 \times 1,3$

(iii) Answer to (c) (ii) $\div \text{mass FA1 used} \times 100\%$ [1]

Evaluation ~~for~~ and specified mass

2(a) Using an electronic balance weigh $W_g(\text{NH}_4)_2\text{SO}_4(s) =$,
 At Times W_g

(b) Measure $X \text{ cm}^3$ of $2,5 \text{ moldm}^{-3} \text{ NaOH}$; which is excess, [1]

(c) Mix and dissolve completely in a flask, / beaker. [1]

$\rightarrow C_3$ Mass of FA1 - Answer to (c)(ii) $\times 100\%$
 Mass of FA1

(d) Plug the flask with the cotton wool soaked in CaCl_2 , [1]

(To trap H_2O ;) [1]

(e) ✓ ✓ Weigh the flask and record the total mass, Y; ✓ *with contents / Record initial vol of syringe.* [1]

(f) Heat the mixture; *to boil for at least 5 minutes* [1]

(To drive off NH_3 ;) [1]

(g) Continue to heat the boiling; *and reweighing* [1]

(h) (Until all NH_3 is driven off) no pungent smell/ mass becomes constant; *Fig. Volume is constant Zg* [1]

(i) Calculate the change in mass, *as by use of Volume of Syringe.* [1]

(Y - ~~X~~) g; [1]

Calculate % NH_4^+ using the formula

$$\frac{\text{Change in mass} / (Y - \cancel{X}) \times \text{Mr}(\text{NH}_4^+)}{\text{Mass}((\text{NH}_4)_2\text{SO}_4 / \text{Wg}) \times \text{Mr}(\text{NH}_3)} \times 100\%;$$

y - z / Vol = y - z [1]

[12] marks]

[Total 10]

Calculate mass of Ammonium ion

(j) $m(\text{NH}_4^+)$ as $\frac{m(\text{NH}_3) \times \text{Mr}(\text{NH}_4^+)}{\text{Mr}(\text{NH}_3)}$ 1

$$\frac{V(\text{NH}_3)}{V_m} \times \text{Mr}(\text{NH}_4^+) \quad \frac{18(Y-Z)}{17}$$

(k) calc % of NH_4^+ as $\frac{18}{17}(Y-Z) \times 100\%$

$$\frac{m(\text{NH}_4)_2\text{SO}_4}{m(\text{NH}_4)_2\text{SO}_4}$$

$$= \frac{18}{17}(Y-Z) \times 100\%$$

1